Code No: 136ER **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
ARTIFICIAL NEURAL NETWORKS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Compare artificial intelligence and neural networks.	[2]
b)	What is boltzmann learning? Explain.	[3]
c)	Explain the learning rate and its significance in NN.	[2]
d)	What is multilayer perceptron? Give example.	[3]
e)	Explain about supervised learning?	[2]
f)	What is cross validation? Explain.	[3]
g)	What are the properties of feature maps?	[2]
h)	What are kohonen's self-organizing maps?	[3]
i)	What is the purpose of dynamical systems? Explain.	[2]
j)	What are called neuro-dynamic systems?	[3]

PART – B

(50 Marks)

2.a) Discuss the Statistical Nature of the Learning Process.
b) Explain the different models of neurons with a neat sketch. [5+5]

OR

3.a) Explain competitive learning with example.
b) List and explain various network architectures. [5+5]

4.a) Explain least mean square algorithm with example.
b) Discuss the relation between Perceptron and Bayes Classifier for a Gaussian Environment. [5+5]

OR

5.a) Derive the learning rule for Back Propagation network.
b) What are the heuristics used in designing Back propagation network? Discuss. [5+5]

6.a) What are the major drawbacks of BP networks? Suggest solutions, to overcome these drawbacks.
b) Explain the distinction between stability and convergence. [5+5]

OR

7.a) Discuss in detail about various network pruning techniques.
b) Write a detail note on Hessian matrix and its operations. [5+5]

8. Explain the architecture and the training algorithm used for Kohonen's SOMs. [10]

OR

9. What is the purpose of learning vector quantization? Explain about adaptive pattern classification. [10]

10. Discuss about the manipulation of attractors as a recurrent network paradigm. [10]

OR

11. Explain the working of a Hopfield network, with a neat sketch of its architecture. [10]

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